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Prismatic F-gauges and Frobenius-liftable schemes

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Abstract

We give a description of prismatic F-gauges on Frobenius liftable schemes in terms of big Fontaine-Laffaille modules. The main conceptual insight is that the Nygaard filtered prismatization of such a scheme is pulled back from the Hodge-filtered de Rham stack of its lift. Along the way we study a notion of p^n-de Rham stacks which provide a lift of prismatic cohomology to $\mathbf{Z}/p^{n+1}$-schemes, with the case $n=1$ being relevant to our study.

Prismatic F-gauges as p-adic motives

Fix a prime p and let k be a perfect field of characteristic p. Work of Bhatt–Lurie and Drinfeld [Bha22, BL22a, BL22b, Dri24] associates to any reasonable k-scheme Y, a p-complete and $W(k)$-linear symmetric monoidal stable ∞-category $F\text{-Gauge}_{/\mathbb{\Delta}}(Y)$, of **prismatic F-gauges on Y**, which one may think of as p-adic étale (mixed) motives on Y.

This category has the following features

1. There is a unit object $\mathbf{1}_Y$ and for each $n \in \mathbf{Z}$ an invertible endofunctor $(-)\{n\}$ so that
$$\mathrm{Ext}^i_{F\text{-Gauge}_{/\mathbb{\Delta}}(Y)}(\mathbf{1}_Y, \mathbf{1}_Y\{n\}) = H^i_{\mathsf{Syn}}(Y, \mathbf{Z}_p(n))$$
where the right hand side is the weight n syntomic cohomology of [BMS19] (already known to Milne in our setting [Mil76]) and can be regarded as p-adic étale motivic cohomology [BM23].
2. If $X \to Y$ is a morphism of schemes then there is a canonical complex $\mathcal{H}_{\mathsf{Syn}}(X) \in F\text{-Gauge}_{/\mathbb{\Delta}}(Y)$ so that $\mathrm{Ext}^i_{F\text{-Gauge}_{/\mathbb{\Delta}}(Y)}(\mathbf{1}_Y, \mathcal{H}_{\mathsf{Syn}}(X)\{n\})$ is closely related to the syntomic cohomology of X via a Leray spectral sequence.
3. There are realization functors from $F\text{-Gauge}_{/\mathbb{\Delta}}(Y)$ to other coefficient systems for various p-adic cohomology theories (e.g. crystalline, prismatic, Hodge-Tate and de Rham along with their filtrations). Moreover, it contains étale local systems as a full subcategory.

Thus $F\text{-Gauge}_{/\mathbb{\Delta}}(Y)$ provides a robust notion of coefficients for syntomic cohomology.

Main motivation: Relation to p-adic (étale) K-theory

A deep motivic property of the complexes $\mathbf{Z}_p(n)$ of [BMS19],[CMM21] is that they provide a filtration on p-adic (étale) K-theory. More concretely there is a spectral sequence
$$H^s_{\mathsf{Syn}}(Y, \mathbf{Z}_p(d)) \Rightarrow K_{2d-s}(Y)^\wedge_p.$$
Thus the category $F\text{-Gauge}_{/\mathbb{\Delta}}(Y)$ contains deep K-theoretic information not just about Y, but also for schemes over Y. Thus it would be desirable to describe $F\text{-Gauge}_{/\mathbb{\Delta}}(Y)$ in certain specific cases.

Cohomological stacks

The association $Y \mapsto F\text{-Gauge}_{/\mathbb{\Delta}}(Y)$ factors through an intermediate association of a p-adic formal stack $Y \mapsto Y^{\mathsf{Syn}}$, the so-called **syntomification of** Y, so that:

1. By definition, $F\text{-Gauge}_{/\mathbb{\Delta}}(Y) \coloneqq \mathcal{D}_{\mathsf{qc}}(Y^{\mathsf{Syn}})$,
2. There is a canonical line bundle $\mathcal{O}\{1\}$ so that $H^i(Y^{\mathsf{Syn}}, \mathcal{O}\{n\}) = H^i_{\mathsf{Syn}}(Y, \mathbf{Z}_p(n))$, where $\mathcal{O}\{n\} = \mathcal{O}\{1\}^{\otimes n}$; in other words the endofunctor $(-)\{n\}$ is just tensoring with the appropriate line bundle.
3. For any morphism $f\colon X \to Y$, the F-gauge $\mathcal{H}_{\mathsf{Syn}}(X) \coloneqq Rf^{\mathsf{Syn}}_* \mathcal{O}_{X^{\mathsf{Syn}}}$ for the associated morphism of p-adic formal stacks $f^{\mathsf{Syn}}\colon X^{\mathsf{Syn}} \to Y^{\mathsf{Syn}}$.

The geometry of Y^{Syn} therefore controls the structure of $F\text{-Gauge}_{/\mathbb{\Delta}}(Y)$.
In fact, the association $Y \mapsto Y^{\mathsf{Syn}}$ is mediated by two further associations of stacks:

1. the association $Y \mapsto Y^{\mathbb{\Delta}}$, the **prismatization** of Y whose quasi-coherent sheaves are prismatic crystals,
2. and the association $Y \mapsto Y^{\mathsf{Nyg}}$ the **Nygaard filtered prismatization**, whose structure morphism to $\mathrm{Spf}(W(k))$ factors over $\mathbf{A}^1/\mathbf{G}_m$, and so the quasi-coherent sheaves carry the Nygaard filtration on prismatic crystals.

The stack $Y^{\mathbb{\Delta}}$ sits inside Y^{Nyg} via two maps $j_{\mathsf{dR}}, j_{\mathsf{HT}}\colon Y^{\mathbb{\Delta}} \hookrightarrow Y^{\mathsf{Nyg}}$, the so-called **de Rham and Hodge-Tate loci**, which in some sense witness the fact that prismatic cohomology is a Frobenius descent of crystalline cohomology.
The stack Y^{Syn} is then defined as
$$Y^{\mathsf{Syn}} \coloneqq \mathrm{coeq}\left(Y^{\mathbb{\Delta}} \underset{j_{\mathsf{HT}}}{\overset{j_{\mathsf{dR}}}{\rightrightarrows}} Y^{\mathsf{Nyg}}\right)$$

Goal

Our goal is to use the geometry of the aforementioned cohomological stacks $Y^{\mathbb{\Delta}}, Y^{\mathsf{Nyg}}$ and Y^{Syn} to describe $F\text{-Gauge}_{/\mathbb{\Delta}}(Y)$ in simple situations. More precisely, we will completely describe all three stacks when Y is smooth and admits a lift to a p-adic formal scheme $\mathfrak{Y}$ over $W(k)$ *along with* a $W(k)$-linear endomorphism $\varphi_{\mathfrak{Y}}$ which lifts the Frobenius on Y. We can also say something for mildly singular schemes with a Frobenius lift to $W(k)$, which we will omit for the sake of brevity.

The Frobenius-liftable case

Let $\mathfrak{Y}/W(k)$ be a smooth p-adic formal scheme. In the sequel we will say that $\mathfrak{Y}/W(k)$ is a δ-scheme if it carries a lift of the Frobenius (this is equivalent to carrying a δ-structure in the smooth case, which is our setting).

Associated to $\mathfrak{Y}/W(k)$ there are three 'de Rham stacks':

- $\mathfrak{Y}^{\mathsf{dR},+}$, which lives over $\mathbf{A}^1/\mathbf{G}_m$ and parametrises Hodge filtered de Rham cohomology,
- $\mathfrak{Y}^{\mathsf{dR}}$ which parametrises de Rham cohomology relative to $W(k)$; it is the specialization of $\mathfrak{Y}^{\mathsf{dR},+}$ to the locus $\mathbf{G}_m/\mathbf{G}_m$.
- $\mathfrak{Y}^{p\text{-}\mathsf{dR}}$, **the p-de Rham stack of $\mathfrak{Y}$**, parametrising a variant of de Rham cohomology where one scales the differential by p; it is the specialization of $\mathfrak{Y}^{\mathsf{dR},+}$ at the point $i_p\colon \mathrm{Spf}(W(k)) \to \mathbf{A}^1/\mathbf{G}_m$ corresponding to the coordinate $t = p$

If $\mathfrak{Y}/W(k)$ is a δ-scheme, then there is a 'divided Frobenius' morphism
$$\frac{F}{p}\colon \mathfrak{Y}^{\mathsf{dR}} \to \mathfrak{Y}^{p\text{-}\mathsf{dR}},$$
which is an isomorphism of stacks.

Strategy

Let $\mathfrak{Y}/W(k)$ be a smooth δ-scheme. Study the syntomification of the special fiber $Y \coloneqq \mathfrak{Y} \otimes_{W(k)} k$ in terms of the structures on the aforementioned de Rham stacks.

The Nygaard filtered prismatization of a Frobenius liftable scheme is pulled back.

In [Sah26b], our first main observation is the following:

Theorem: pulling back the Nygaard filtered prismatization

Theorem (S. [Sah26b]) Let $\mathfrak{Y}/W(k)$ be a smooth δ-scheme and $Y \coloneqq \mathfrak{Y} \otimes_{W(k)} k$. Then there is a pullback square
$$\begin{array}{ccc} Y^{\mathsf{Nyg}} & \longrightarrow & \mathfrak{Y}^{\mathsf{dR},+} \\ \downarrow & & \downarrow \\ k^{\mathsf{Nyg}} & \longrightarrow & \mathbf{A}^1/\mathbf{G}_m \end{array}$$
This pullback is functorial in the δ-scheme $\mathfrak{Y}/W(k)$.

Proof sketch: The first proof of this result, due to the author, hinges on an observation of [BMS19] (going back to Illusie) which says that for a δ-scheme $\mathfrak{Y}/W(k)$ the Nygaard filtration on the crystalline cohomology of the special fiber Y is just the tensor product of the Hodge filtration and the p-adic filtration on the de Rham cohomology of $\mathfrak{Y}/W(k)$. To lift this statement, which is at the level of cohomology, to one for stacks, we use a notion of *affineness in the derived sense* due originally to Toën [Toë06] and recast into modern form in [MM25].

More precisely, we show in [Sah25] the following theorem (stated a bit heuristically to avoid technicalities)

Theorem: de Rham affineness

Theorem(S. [Sah25]): Let Y be any affine scheme over k. Then Y^{Nyg} viewed as a stack over k^{Nyg} is isomorphic to the relative spectrum of the Nygaard filtered prismatic cohomology of Y. This isomorphism is functorial in Y.

A similar result is shown for the stack $\mathfrak{Y}^{\mathsf{dR},+}$ as a stack over $\mathbf{A}^1/\mathbf{G}_m$ in [Sah26b]. The functorial affineness which we dub **de Rham affineness**, then formally determines the displayed pullback square.

Making this theory work involved making sense of **non-connective animated rings over formal stacks**. This work is carried out in [Sah26a].

Gauges as filtered $\mathcal{D}$-modules

Let $\mathfrak{Y}/W(k)$ be a smooth δ-scheme. $\mathcal{D}^{\mathsf{crys}}_{\mathfrak{Y}}$ be the sheaf of crystalline differential operators i.e. the universal enveloping algebroid of the tangent bundle of $\mathfrak{Y}$. Equip it with the **Nygaard filtration** $\mathrm{Fil}_{\mathsf{Nyg}}(\mathcal{D}^{\mathsf{crys}}_{\mathfrak{Y}})$, defined as the tensor product of the order filtration and the p-adic filtration.

An immediate pay off of the previous theorem is the following

Theorem: Gauges as filtered $\mathcal{D}_{\mathfrak{Y}}$-modules

Theorem(S. [Sah26b]) Let $\mathfrak{Y}/W(k)$ be a smooth δ-scheme with special fiber Y. There is an equivalence
$$\mathcal{D}_{\mathsf{qc}}(Y^{\mathsf{Nyg}}) \simeq \mathrm{FilMod}^{\mathsf{Inilp}}_{p\text{-comp}}\left(\mathrm{Fil}_{\mathsf{Nyg}}(\mathcal{D}^{\mathsf{crys}}_{\mathfrak{Y}})\right),$$
where the right-hand side consists of derived p-complete filtered complexes with locally nilpotent p-curvature modulo p.

Differential F-gauges

The Rees construction applied to $\mathrm{Fil}_{\mathsf{Nyg}}(\mathcal{D}^{\mathsf{crys}}_{\mathfrak{Y}})$, along with the pushout description of Y^{Syn}, and with the interactions of the de Rham stacks described previously, one gets the following concrete description of prismatic F-gauges on a Frobenius liftable scheme:

Theorm: Differential F-gauge description of Prismatic F-gauges

Theorem(S. [Sah26b]) Let $\mathfrak{Y}/W(k)$ be a smooth δ-scheme and Y the special fiber. Objects of $\mathcal{D}_{\mathsf{qc}}(Y^{\mathsf{Syn}})$ correspond to graded, derived p-complete modules $M^\bullet$ over
$$\mathrm{Rees}\left(\mathrm{Fil}_{\mathsf{Nyg}}(\mathcal{D}^{\mathsf{crys}}_{\mathfrak{Y}})\right)$$
with locally nilpotent p-curvature modulo p, and equipped with an isomorphism
$$F^*M^\bullet\big|_{t=p} \;\simeq\; M^\bullet\big|_{t=1}.$$
On vector bundles the F-pullback is the F-transform of Ogus and Shiho.

This is a relative 'big Fontaine–Laffaille' theory, in the sense of [TVX25]: there is no restriction on Hodge–Tate weights.

Recovering a result of Ogus on the comparison between prismatic crystals and p-connections

In recent work [Ogu24], Ogus has given a description of prismatic crystals in terms of p-connections. We are able to recover a stacky version of this result.

Theorem: recovering Ogus

Theorem(Ogus [Ogu24], [Sah26b]) Let $\mathfrak{Y}/W(k)$ be a smooth δ-scheme with special fiber Y. Then there is an equivalence of p-adic formal stacks over $\mathrm{Spf}(W(k))$
$$\mathfrak{Y}^{p\text{-}\mathsf{dR}} \simeq Y^{\mathbb{\Delta}}.$$
In particular there is an equivalence of stable ∞-categories
$$\mathrm{Crys}_{/\mathbb{\Delta}}(Y) \simeq \mathcal{D}_{\mathsf{qc}}(\mathfrak{Y}^{p\text{-}\mathsf{dR}})$$
where the left hand side is prismatic crystals and the right hand side is just modules over a sheaf of p-differential operators.

A mod p^n lift of prismatic cohomology and a mystery

Let $\mathfrak{Y}/W(k)$ be any smooth p-adic formal scheme. We have defined the p-de Rham stack $\mathfrak{Y}^{p\text{-}\mathsf{dR}}$.

p^2-crystallinity of the p-de Rham stack

For any smooth p-adic formal scheme $\mathfrak{Y}/W(k) \mapsto \mathfrak{Y}^{p\text{-}\mathsf{dR}}$ factors through the mod p^2-fibre i.e. $\mathfrak{Y}/W(k) \mapsto \mathfrak{Y} \otimes_{W(k)} W(k)/p^2$.

As a consequence there is a 'new' cohomology theory on $W(k)/p^2$-schemes, valued in (derived) p-complete $W(k)$-modules. Explicitly, this sends any smooth $W(k)/p^2$-scheme Y, to the cohomology $Y \mapsto H^*(Y^{p\text{-}\mathsf{dR}}, \mathcal{O})$. This assignment depends on the scheme Y and **not just the special fibre**.

In fact for any $n > 0$ there is a p^n-de Rham stack

Theorem/ Expectation: mod p^n-twists of prismatic cohomology

For every $n > 0$ there is a 'new' cohomology theory on $W(k)/p^{n+1}$-schemes with values in (derived) p-complete $W(k)$-modules. Explicitly for every smooth $Y \in \mathrm{Sch}_{W(k)/p^{n+1}}$ there is a stack
$$Y^{p^n\text{-}\mathsf{dR}} \longrightarrow \mathrm{Spf}(W(k))$$
so that the cohomology theory is $Y \mapsto H^*(Y^{p^n\text{-}\mathsf{dR}}, \mathcal{O})$. This cohomology theory is uniquely characterized as a functorial lift of the p^n-de Rham complex:
$$R\Gamma(Y^{p^n\text{-}\mathsf{dR}}, \mathcal{O}) \otimes^{\mathbf{L}} W(k)/p^{n+1} \simeq R\Gamma\big(Y,\, \mathcal{O}_Y \xrightarrow{p^n d} \Omega^1_Y \xrightarrow{p^n d} \Omega^2_Y \to \cdots\big).$$
When Y is a mod p^{n+1} Frobenius lift of its special fibre Y_0, then this cohomology theory is **just the prismatic cohomology of Y_0. In general this theory does not depend only on the special fibre**.

The author would like to know if there is a good reason for these deformations to exist.

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